

Vehicle-BD: A Benchmark Dataset of Bangladeshi Local Vehicles

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Abstract. With the increasing power of technology, smart traffic control systems have become a vital research domain for every country. For any Intelligent Transport System (ITS), vehicle detection, localization, and classification are the main tasks. Large-scale databases are a key component of data-driven, sophisticated object detection models. The use of well-labeled datasets has considerably enhanced the efficacy and accuracy of many different object identification tasks. Building up a dataset of vehicles is hence the initial stage in detecting vehicles on Bangladeshi roads. Considering the scarcity of datasets for local vehicles in Bangladesh, we have proposed a new dataset called the "Vehicle BD" dataset for vehicle localization and categorization. This dataset is comprised of a total of 12,413 images. We have considered nine types of local vehicles: bus, truck, car, CNG, easy bike, motorcycle, bicycle, rickshaw, and van. Data labeling has been done by using "Labelling". For privacy purposes, we have also blurred the faces in the images. This dataset can be a benchmark for research in the context of the Bangladesh road scenario.

Keywords: vehicle detection, vehicle classification, local vehicles, dataset, intelligent traffic system.

1 Introduction

Identification, localization, and type classification are essential elements of an intelligent traffic system because they allow for detecting cars in images and indicating their location via bounding boxes [9].

Automated vehicle identification and classification have a variety of uses in both the private and governmental sectors because of the enormous expansion of computer vision technology. It enables low-cost, ongoing, and comprehensive traffic monitoring that has an impact on a nation's economic development [19]. This field of research has a huge impact on building an intelligent traffic system [9]. In a developing nation like Bangladesh, traffic infractions and accidents on the road are increasingly frequent. Numerous factors, including rash driving, speeding, forceful overtaking, breaking traffic laws, prolonged periods of constant driving, impairment by drugs or alcohol, hazardous roads, a deficient traffic management system, and a lack of public awareness, have been identified by researchers as contributing to the rising number of traffic anomalies [7].

However, these kinds of automatic systems require a scaleable, standard, and realistic dataset. Any data-driven approach depends largely on the quality of the dataset. Therefore, it is undoubtedly obvious to build a standard dataset that will contain images of frequently used local vehicles on the roads of Bangladesh. It will not only aid in the development of the intelligent traffic system but will also point to further research opportunities. The existing benchmark datasets [15,13,22,18] don't match Bangladesh's road situation. As a result, any detection or classification system that is developed using this dataset would have low performance. That is why we made the effort to conduct this research by compiling a realistic, standard dataset of Bangladeshi local vehicles.

The contributions of our work are listed as follows:

- We have collected real images from the roads of Bangladesh.
- Three major cities were covered to collect the photos to make it more challenging and realistic.
- A Total of nine types of mostly used local vehicles are considered.
- We have taken pictures from different viewpoints to make it more diverse.
- Labeled the data and augmented it considering different situational effects

The remainder of the paper is structured as follows: Section 2 discusses related efforts in this field; Section 3 provides information on the benchmark dataset; Section 4 explains the dataset's significance and influence across domains; and Section 5 contains the concluding remarks.

2 Related Works

Creating an acceptable dataset for vehicle classification is a gruesome task. There have been some recent efforts to create a comprehensive dataset all over the world. Among the significant datasets, Mohsin Ali et al. in [1] have made a collection of 3847 images that were created from high-resolution (1920 1080) films recorded from camera units placed on a roadway at various angles and at varying frame rates. The dataset contains 48 classes of various car models that are labeled in 48 distinct folders. This increased the difficulty of this dataset while assisting in the collection of photographs showing a real-world scenario.

Sebastien Drouyer in [3] presented a sizable dataset (VehSat) for vehicle detection on satellite images. They have gathered 4544 crops of satellite photos in 8 different places from 4 different satellites. 36851 automobiles were annotated overall in a range of settings and resolutions. The vehicle density varied significantly across the dataset: there are 2202 automobiles per square kilometer in Boston, USA, compared to 760 in Khartoum, Sudan. The number of cars per crop ranged from 0 to 124, with an average of 7.5 and a standard deviation of 10.1.

In the context of our work, there is one related research that has been done in [17] by Shaira Tabassum et al. The authors have introduced a new dataset called the "Poribohon-BD" dataset to classify vehicles in Bangladesh. Two sources are used to get photographs of the vehicles: social media and smartphone cameras.

The collection has 9058 pictures of 15 native species that have been labeled and annotated Bangladeshi automobiles, including buses, motorcycles, three-wheelers, wheelbarrows, trucks, etc. Strategies for data augmentation are applied to maintain an equal number of photos for each type of vehicle. LabelImg is used to label the images. Besides, human faces have also been obscured to protect confidentiality and privacy. However, the main drawback of this dataset is that the pictures are not collected from real roads. Some of it is downloaded from the internet, and some of it is from foreign vehicles. Therefore, it cannot be a benchmark dataset for Bangladeshi vehicle detection as the images are not applicable in real road scenarios.

Hai Wang et al. in [20] have introduced a vehicle reidentification (REID) dataset "V2I CARLA" based on an automated driving simulator, successfully identifying the same vehicle while the shared features are few from various angles. This article suggests a method for car REID using the pyramid approach. The pyramid method tends to obtain multiscale features of the vehicle under various cameras, capturing small feature differences between automobiles with a similar look. They have gathered information on 153 distinct car types and colors. The vehicle that is visible in the captured photograph is then excised from the image, and the cutout vehicle image is labeled with the camera ID and vehicle ID information. 1500 dual-view photos with various IDs are included in the collection, of which 1045 serve as training images and 455 as test images.

Zhiming Luo et al. in [12] focused on categorizing and locating motor vehicles from a single video frame, and the authors introduced the "Miovision traffic camera dataset" (MIO-TCD) in this context. A total of 786,702 photos make up the MIO-TCD dataset, including 648,959 lower-resolution photographs of cropped automobiles and 137,743 high-resolution video frames showing several vehicles. These photos were taken by hundreds of traffic cameras that were placed all over Canada and the United States at various hours of the day and throughout the seasons. These pictures are typical of the visual data taken in both urban and rural traffic, and they span a wide variety of difficulties. The collection is divided into two sections: the "localization dataset," which includes 137743 complete video frames with bounding boxes around traffic items, and the "classification dataset," which includes 648,959 crops of traffic objects from the 11 classes.

A large-scale dataset for vehicle re-identification (ReID) was created by Peng Wang et al. in [21] that contains 137k photos of 13k vehicle instances that were photographed by UAV-mounted cameras. They finally amassed 350 pairs of video clips totaling 34 hours in duration (about 3 minutes per clip), with more than 200 man-hours of UAV flight and video recording. Every 0.5 seconds, they sampled frames from each clip to get a total of 25, 2000 images.

To get a closer view of vehicles, there have been some research works for number plate recognition, where several datasets were found that contained images captured in close proximity. Naaman Omar Yaseen et al. in [23] provided a brand-new dataset for vehicle photos from three provinces in northern Iraq, dubbed North Iraq-Vehicle Photos (NI-VI): Duhok, Erbil, and Sulaimani. This dataset contains 1500 pictures. To create a realistic dataset of the vehicle photos,

they were collected in real time using handheld cameras. The primary contribution of this work is the development of a new dataset for Arabic-font license plates on automobiles in northern Iraq under various challenging settings. Three types of image categories are present in the dataset: rotated, scaled, and translated images. Images have 4288 x 2848 and 5184 x 3456 resolutions, respectively.

Lei Lu et al. have conducted research on fine-grained vehicle categorization in [11]. To advance the development of vision-based research on the vehicle, and specifically for the purpose of fine-grained vehicle categorization, they suggested a picture collection called Frontal-103. An in-depth examination of Frontal-103's present configuration, which includes 65,433 web-nature photos and 1,759 fine-grained car models from 103 different vehicle brands, is provided in this report. Frontal-103 is superior to other cutting-edge vehicle picture datasets in terms of scale, diversity, correctness, and fine-grained level, aside from the particular viewpoint and vehicle hierarchy. Mourad A. Kenk et al. in [8] introduced DAWN, a brand-new dataset (benchmark) made up of actual photographs taken in various unfavorable weather situations. This dataset highlights both a vast range of traffic flow and a diverse traffic environment (urban, highway, and freeway). The DAWN dataset consists of 1,000 photos taken in actual traffic situations and is broken down into four categories of weather: fog, snow, rain, and sandstorms. The dataset has object-bounding boxes annotated for situations involving autonomous vehicles and video surveillance. This information aids in the interpretation of the effects that bad weather has on the effectiveness of vehicle detection systems.

More recently, Xinchun Liu et al. in [10] have offered the vehicle reidentification (VeRi) benchmark dataset, a sizable dataset set in the context of urban surveillance. 619 vehicles, 7,021 tracks, and 40,395 bounding box photos are included in the VeRi dataset. This increases the scalability of their dataset for the vehicle re-id task. Furthermore, each vehicle is filmed by 218 cameras at various angles, circumstances of lighting, and resolutions to ensure a high recurrence rate for vehicle recognition. Finally, they proposed a baseline that incorporates deep neural network-extracted high-level semantic information with color, texture, and other visual signals, utilizing VeRi, to evaluate six rival vehicle Re-Id techniques.

3 Dataset

Unfortunately, due to focusing more on quantity than quality, some of the recently introduced datasets, such as Poribohon-BD and Deshi-BD, lack usability. In many cases, these datasets do not reflect actual road scenarios. Additionally, there are a few categories of vehicle images that are hardly or no longer seen on Bangladeshi roads. Moreover, datasets include images collected from social media, often providing misleading information about the actual situation. Furthermore, a lack of proper augmentation of each image creates detrimental effects on building a robust intelligent traffic management system. While building the "Vehicle-BD" dataset, these considerations have been carefully handled.

Table 1: Data Collection Process Specification

Sl.No.	Specification	Details
1	Data Type	2D-RGB image (JPG)
2	Data Format	Raw Digital Images(.jpg)
3	Data Acquiring Device	Smartphone (Huawei Nova 3i)
4	Camera Quality	16MP, f/2.2
5	Primary Image Size	960x1280px
6	Original Aspect Ratio	3:4
7	Total Vehicle Types	9
8	Time of Day	Morning and Afternoon
10	Location	Dhaka, Khulna, and Bogura
11	Availability	https://tinyurl.com/bddkayyp

Decades ago, it was hard to find a large and scalable dataset of local Bangladeshi vehicles to perform any type of research. However, the time has changed, and researchers are trying to gather images from different sources to form a large dataset of Bangladeshi vehicles.

The dataset creation process has been divided into three important phases, which are data collection, data augmentation, and data annotation. All the phases have been performed sequentially.

3.1 Data Collection

Data collection is one of the most challenging phases of building a proper image dataset. Collecting local vehicle images from the actual roads adds an extra layer of difficulty to this phase. Table 1 summarizes the data collection phase details and provides a reference to the dataset.

Images were originally captured using a Huawei Nova 3i smartphone during the morning and afternoon hours of the day. In total, nine different vehicles (bus, truck, car, CNG, easy bike, motorcycle, bicycle, rickshaw, and van) are considered the most frequently seen vehicles on the roads, and images of them were collected from the roads of three important cities in Bangladesh, which are Dhaka, Khulna, and Bogura, to add diversity to the dataset. All the images were captured in 960x1280px RGB format.

Fig. 1 illustrates the count of images collected per class. From the figure, it can be noticed that all nine types of vehicles have almost the same number of images. This signifies that there are no noticeable biases among the types of vehicles in the dataset. Additionally, a mixed category of images has been introduced, where each image contains multiple types of vehicles. Fig. 2 represents the images that are collected from different roads.

3.2 Data Augmentation

Collecting vehicle images in every possible orientation and background is not always a feasible idea. Moreover, there can be situations where the camera quality

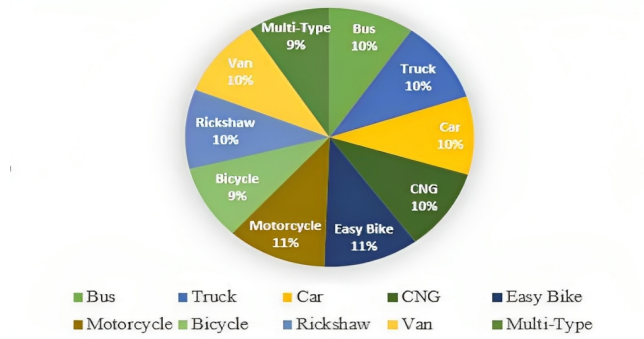


Fig. 1: Distribution of images per class after data collection phase

Table 2: Specifics of Data Augmentation

Sl.No.	Modification	Details
1	Resize	Every image was reduced to 750×1000 pixels with a comparable aspect ratio.
2	Rotation	Randomly selected photos have been rotated between -10 to 10 degrees.
3	Blur	Upto 1.25px gaussian blur has been included in the randomly chosen images.
4	Noise	A maximum of 4% gaussian noise is added to the arbitrarily selected images.

is degraded and may not capture clear images. To address this situation, the original captured data has been augmented through a series of operations. Table 2 presents the details of the augmentation operations.

The resize operation is performed on images to scale down the size of the images for ease of use. Rotation has been added to make the dataset more robust so that the developed system can achieve scale-invariance capability. Some blurriness is introduced to imitate the situation that can be experienced when a vehicle is moving fast. Finally, some added salt and pepper noise can help create the effect of a rainy day. Table 3 displays the total number of photos in each class in the collection. It is evident that each class has grown in size by more than ten times following the augmentation. It displays the variety of photos included in the collection. The 12,413 photos in the final dataset feature nine distinct car makes. Fig. 3 presents that the augmentation doesn't change the overall distribution of the dataset. All the vehicles have almost the same distribution as before.

3.3 Data Annotation

For recognizing a specific object in an image, annotating the object in the training dataset is crucial. At this stage of dataset preparation, all the images have been manually annotated. To do so, "labeling", open-source software is used.



Fig. 2: Raw images of Vehicle-BD dataset

The annotated file is in XML format. Fig. 4 presents the images with the annotated bounding boxes. There is one sample of annotated images from each vehicle type and one from the multi-class. In addition, Fig. 5 shows an example annotated file. In the file, the "size" tag groups the information (width, height, and depth) on the image file. The bounding box information is inside the "object" tag. "xmin", "ymin", "xmax", and "ymax" tags indicate the 2D coordinate values of the four corners of the bounding box in an image. The name of the object that is present in the image is specified in the "name" tag.

4 Significance and Impact

Bangladesh is a densely populated developing country that has a wide range of research opportunities in many sectors. The road and transport sector is one of them. From 2021 to 2022, more than 100,000 motor vehicles have been registered, according to the data provided by the Bangladesh Road and Transport Authority (BRTA). An increasing trend is observed in the registration of buses, private cars, jeeps, minibuses, autorickshaws, and motorcycles. This new addition of vehicles each year creates new challenges for the traffic management authority to properly manage rules and regulations on the road. Thus, researchers are focusing on solving these challenges with more determination than ever. In this regard, a rich dataset of Bangladeshi local vehicles can significantly improve the future prospects of research in this sector. Some noteworthy use cases of the dataset introduced in this article are:

Table 3: Per Class Images in Dataset

Sl.No.	Vehicle Class	Initial Dataset	Augmented Dataset
1	Bus	104	1,220
2	Truck	112	1,264
3	Car	106	1,121
4	CNG	111	1,284
5	Easy Bike	116	1,436
6	Motorcycle	121	1,367
7	Bicycle	101	1,147
8	Rickshaw	108	1,220
9	Van	106	1,226
10	Multi-Type	100	1,128
	Total	1085	12,413

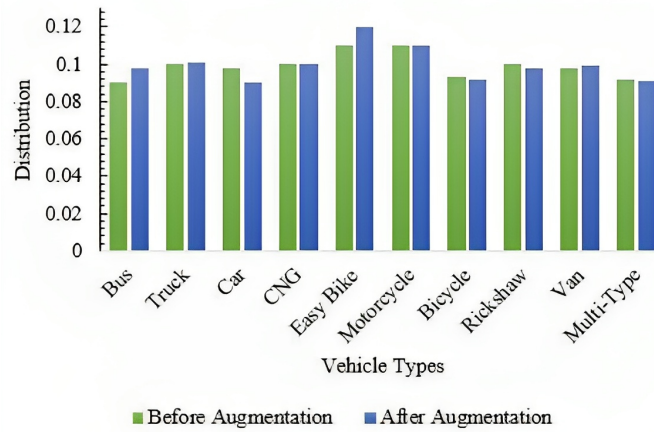


Fig. 3: Distribution of images per class after data annotation phase

- Researchers interested in vehicle identification and segmentation will be the main beneficiaries of this dataset [14].
- This dataset can be used in building models for automobile detection, classification, and segmentation.
- Particularly, for research towards a better traffic management system, this dataset can be a gold resource [5].
- Although this dataset has the nine most common types of vehicles on the road, extending this dataset is easy to do and can help in developing a more robust system.
- Identification of unfit vehicles, unauthorized vehicle detection, highway toll collection, garage space record tracking, and traffic information collection are some promising areas where this dataset can be used to build a proper system.
- Research for beginning the use of autonomous vehicles on Bangladeshi can be greatly improved with this dataset.



Fig. 4: Images from Vehicle-BD dataset after augmentation and annotation

- Information regarding the necessity of introducing new laws for managing traffic in an efficient way can be collected with the help of a monitoring system built using this dataset.

The table 4 presents contrastive information regarding the current datasets, which are used for vehicle recognition. Here, the largest one is the Waymo dataset, with around 12 million images. However, it only has two types of vehicle information. Cityscapes is the second-largest dataset, with 25000 images and 6 vehicle classes. There are also KITTY and ApolloScape datasets with five and six vehicle category classes. These datasets have semantic, fine-coarse, LiDAR box-camera box, and 3D bounding box annotations, which limit the potential of the dataset to be used in a wide variety of applications.

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Vehicle-BD covers these limitations and has been prepared to provide images of real-life road conditions. Furthermore, some augmentation has also been done to make images look more realistic and weather-invariant.

```

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  </source>
  <size>
    <width>750</width>
    <height>1000</height>
    <depth>3</depth>
  </size>
  <segmented>0</segmented>
  <object>
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    <pose>Unspecified</pose>
    <truncated>0</truncated>
    <difficult>0</difficult>
    <occluded>0</occluded>
    <bndbox>
      <xmin>133</xmin>
      <xmax>583</xmax>
      <ymin>316</ymin>
      <ymax>662</ymax>
    </bndbox>
  </object>
</annotation>

```

Fig. 5: Manual image annotation file

Table 4: Comparison with Other Datasets

SL. No.	Name	Number of Images	Number of Vehicle Classes	Annotation Type
1	ApolloScape [6]	701	6	Semantic Annotation
2	KITTI [4]	7481	5	3D Bounding Box
3	Cityscapes [2]	25000	6	Fine-Coarse Annotation
4	Waymo [16]	Around 12 million	2	LiDER Box - Camera Box Annotation
5	Poribohon-BD [17]	9058	15	2D Bounding Box
6	Vehicle-BD	12413	9	2D Bounding Box

As a whole, the dataset has more than 12,000 images of the nine most common vehicles in the Bangladeshi transport sector, making it the largest Bangladeshi local vehicle image dataset available till now.

5 Conclusion

In this study, we introduce a brand-new dataset called Vehicle-BD, which includes data on nine different kinds of locally accessible automobiles in Bangladesh. The 12,413 are all appropriately labeled. This information can be used to create data-driven applications for traffic volume measurement and automatic traffic monitoring systems. Our main goal was to create a benchmark dataset and concentrate on Bangladesh's roads. There are no realistic photos of nearby automobiles in the current dataset. This dataset may reveal fresh lines of inquiry. We intend to include more local automobiles and more complicated road circumstances in the research's future extension.

References

1. Mohsin Ali, Muhammad Atif Tahir, and Muhammad Nouman Durrani. Vehicle images dataset for make and model recognition. *Data in brief*, 42, 2022.
2. Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The cityscapes dataset for semantic urban scene understanding. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3213–3223, 2016.
3. Sébastien Drouyer. Vehsat: A large-scale dataset for vehicle detection in satellite images. In *IGARSS 2020-2020 IEEE International Geoscience and Remote Sensing Symposium*, pages 268–271. IEEE, 2020.
4. Andreas Geiger, Philip Lenz, Christoph Stiller, and Raquel Urtasun. Vision meets robotics: The kitti dataset. *The International Journal of Robotics Research*, 32(11):1231–1237, 2013.
5. Syed Nahin Hossain, Md Zahim Hassan, and Md Masum Al Masba. Automatic license plate recognition system for bangladeshi vehicles using deep neural network. In *Proceedings of the International Conference on Big Data, IoT, and Machine Learning: BIM 2021*, pages 91–102. Springer, 2022.
6. Xinyu Huang, Xinjing Cheng, Qichuan Geng, Binbin Cao, Dingfu Zhou, Peng Wang, Yuanqing Lin, and Ruigang Yang. The apolloscape dataset for autonomous driving. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 954–960, 2018.
7. Md Ashraful Islam and Yousuf Dinar. Evaluation and spatial analysis of road accidents in bangladesh: an emerging and alarming issue. *Transportation in developing economies*, 7:1–14, 2021.
8. Mourad A Kenk and Mahmoud Hassaballah. Dawn: vehicle detection in adverse weather nature dataset. *arXiv preprint arXiv:2008.05402*, 2020.
9. Yangxin Lin, Ping Wang, and Meng Ma. Intelligent transportation system (its): Concept, challenge and opportunity. In *2017 IEEE 3rd international conference on big data security on cloud (bigdatasecurity), IEEE international conference on high performance and smart computing (hpsc), and IEEE international conference on intelligent data and security (ids)*, pages 167–172. IEEE, 2017.
10. Xinchun Liu, Wu Liu, Huadong Ma, and Huiyuan Fu. Large-scale vehicle re-identification in urban surveillance videos. In *2016 IEEE international conference on multimedia and expo (ICME)*, pages 1–6. IEEE, 2016.

11. Lei Lu, Ping Wang, and Hua Huang. A large-scale frontal vehicle image dataset for fine-grained vehicle categorization. *IEEE Transactions on Intelligent Transportation Systems*, 23(3):1818–1828, 2020.
12. Zhiming Luo, Frederic Branchaud-Charron, Carl Lemaire, Janusz Konrad, Shaozi Li, Akshaya Mishra, Andrew Achkar, Justin Eichel, and Pierre-Marc Jodoin. Mio-tcd: A new benchmark dataset for vehicle classification and localization. *IEEE Transactions on Image Processing*, 27(10):5129–5141, 2018.
13. T Nathan Mundhenk, Goran Konjevod, Wesam A Sakla, and Kofi Boakye. A large contextual dataset for classification, detection and counting of cars with deep learning. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part III 14*, pages 785–800. Springer, 2016.
14. Md Sazedur Rahman, Md Zahim Hassan, Syed Nahin Hossain, Noor Masrur, and Jakaria Rabbi. Bangladeshi local vehicle recognition with a comprehensive dataset using transfer learning techniques. In *2022 4th International Conference on Sustainable Technologies for Industry 4.0 (STI)*, pages 1–6. IEEE, 2022.
15. Sebastien Razakarivony and Frederic Jurie. Vehicle detection in aerial imagery: A small target detection benchmark. *Journal of Visual Communication and Image Representation*, 34:187–203, 2016.
16. Pei Sun, Henrik Kretzschmar, Xerxes Dotiwalla, Aurelien Chouard, Vijaysai Patnaik, Paul Tsui, James Guo, Yin Zhou, Yuning Chai, Benjamin Caine, et al. Scalability in perception for autonomous driving: Waymo open dataset. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2446–2454, 2020.
17. Shaira Tabassum, Sabbir Ullah, Nakib Hossain Al-Nur, and Swakkhar Shatabda. Poribohon-bd: Bangladeshi local vehicle image dataset with annotation for classification. *Data in brief*, 33, 2020.
18. Franklin Tanner, Brian Colder, Craig Pullen, David Heagy, Michael Eppolito, Veronica Carlan, Carsten Oertel, and Phil Sallee. Overhead imagery research data set—an annotated data library & tools to aid in the development of computer vision algorithms. In *2009 IEEE Applied Imagery Pattern Recognition Workshop (AIPR 2009)*, pages 1–8. IEEE, 2009.
19. Floris van Ruth. *Traffic intensity as indicator of regional economic activity*. Statistics Netherlands, 2014.
20. Hai Wang, Xin Yuan, Yingfeng Cai, Long Chen, and Yicheng Li. V2i-carla: a novel dataset and a method for vehicle reidentification-based v2i environment. *IEEE Transactions on Instrumentation and Measurement*, 71:1–9, 2022.
21. Peng Wang, Bingliang Jiao, Lu Yang, Yifei Yang, Shizhou Zhang, Wei Wei, and Yanning Zhang. Vehicle re-identification in aerial imagery: Dataset and approach. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 460–469, 2019.
22. Gui-Song Xia, Xiang Bai, Jian Ding, Zhen Zhu, Serge Belongie, Jiebo Luo, Mihai Datcu, Marcello Pelillo, and Liangpei Zhang. Dota: A large-scale dataset for object detection in aerial images. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3974–3983, 2018.
23. Naaman Omar Yaseen, Salim Ganim Saeed Al-Ali, and Abdulkadir Sengur. Development of new anpr dataset for automatic number plate detection and recognition in north of iraq. In *2019 1st International Informatics and Software Engineering Conference (UBMYK)*, pages 1–6. IEEE, 2019.